

BEFORE

As settlements became more complex, trade and ties between communities grew and material culture became more developed.

BEFORE METAL

Flint was used to make the hardest and sharpest tools and weapons for thousands of years. From simple chopping devices, humans went on to produce tools for particular tasks, including scrapers, cutters, and hand-axes << 17.

PERMANENT SETTLEMENTS

As agriculture spread throughout southwestern Asia after 8000 BCE, settlements became more permanent. With permanent settlement came closer ties to the land and to a specific territory and its resources.



OBSIDIAN

EARLY TRADE

Most communities exchanged commodities such as grain, building materials, and toolmaking stone. As a result, trade networks and relationships grew. The village of Çatalhöyük in Turkey, for example, specialized in trading obsidian, a highly prized volcanic glass used for making tools, particularly sharp knives.

RITUAL AND WORSHIP

Both ritual and social life soon became more elaborate. At Çatalhöyük, what appear to be family shrines, where people commemorated their ancestors << 32 and worshipped a fertility goddess, have been excavated, revealing distinctive goddess figurines made of clay.



GODDESS FIGURINE

Precious Metal

Humans had made tools out of stone, bone, and wood for thousands of years. The advent of copperworking around 8,000 years ago was the beginning of a long association with metals, and a significant watershed in human history, which led to even further innovation.

It seems that humans have always had a fascination with rare and beautiful materials,

from exotic shells used as jewelry to shiny metals. Copper occurs both in its native form (uncombined

with other elements and so needing little processing) or in the form of ore (a rock or mineral from which the metal can be extracted). Native copper is very malleable, with a distinctive reddish color. When it was first found, it was mainly used to make shiny ornaments. The earliest copper objects were hammered into shape to form crude axes and beads—examples of

these dating back to the 6th millennium BCE have been found in Turkey and the mountains of

Iran. Copper ores are relatively common around the Mediterranean. Those found in surface outcrops are easy to identify by

their distinctive green color. It was the discovery of copper smelting (heating the ore with charcoal to about 2,200°F [1,200°C] to extract the metal) that opened the door to a world of innovation, and the development of stronger and more practical items. This crucial discovery is believed to have been made at some point before 5000 BCE and may, theoretically, have

resulted from ore being dropped into a hearth by accident and reacting with hot embers. The earliest known use of gold and silver dates to a similar period. Gold was found in alluvial deposits in or along streams—which meant it did not have to be mined—and was valued for decorative and ritual purposes including grave ornaments (see right).

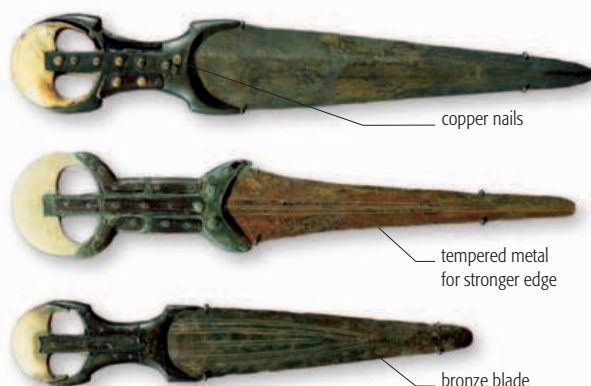
The first metalworkers smelted copper in open fires. They soon began to use holes lined with clay, then crucibles (containers designed to withstand the heat of the fire), to produce the metal more efficiently.

Trading up

The use of copper not only stimulated technological advances, but also led to significant cultural developments. As copper ore outcrops are patchily distributed, copper items and metal ingots of standard shapes and sizes became valued as tradable items. Culturally advanced in other ways, lowland Mesopotamia had no native metal or ore, which meant that both copper and gold, the two most prestigious ornamental metals, were imported from Anatolia (modern Turkey) and the Iranian Plateau. The traffic in these materials grew quickly, and growing lowland Mesopotamian towns began to exchange grain and other commodities for imported artifacts and ornaments. This trade came to involve both overland travel and transport along the Tigris and Euphrates rivers, where rafts made of wood and inflated goatskins probably came into use by 3000 BCE.

Metalworking reaches Europe

The earliest copper mines in Europe have been found at Ai Bunar in southern Bulgaria, and date to around 5100 BCE. A large network of exchange



Egyptian daggers

These daggers from late predynastic Egypt date from c. 3000 BCE. They are made from bronze with copper nails and ivory detail on the handle. As technology developed, tools and weapons that had previously been cut from stone were made in metals, which were lighter and more pliable.

Metal moves west

Metalworking started in Anatolia and Mesopotamia (modern-day Turkey and Iraq) in around 5000 BCE, and spread westward and northward, reaching northern Europe by 2000 BCE. Its spread was facilitated by widespread trade, which was growing during this period.

KEY

- Metalworking 5000–4000 BCE
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- Site
- Sporadic trade route
- Spread of metalworking

